

Rakesh Choudhary

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Professional Summary

Backend-focused engineer with experience building AI-assisted applications, realtime systems, and scalable backend services using Node.js, LangChain, OpenAI APIs, and cloud infrastructure. Skilled in workflow automation, API orchestration, and production deployments.

Professional Experience

Software Engineer | Monkhub Innovations | Gurgaon, India Aug 2025 - Present

- Architected backend systems for blockchain token swaps and bridge workflows, orchestrating async transactions and validating integrity via automated tasks
- Designed secure Fireblocks API integrations and Web3 wallet authentication, ensuring high reliability for critical financial operations
- Led backend optimization initiatives, improving API response times by 30% through advanced MongoDB aggregation pipelines and query caching

Full Stack Developer | Code B Solutions Pvt. Ltd. | Mumbai, India Sep 2024 - May 2025

- Led the development of high-performance media streaming and logistics platforms deployed across 1000+ transit vehicles, utilizing Node.js, ffmpeg, and GCP compute infrastructure
- Formulated a custom Node.js/ffmpeg media pipeline to capture and broadcast live transit feeds with audio at 30 FPS, enhanced for variable network bandwidths, and reduced stream latency by 20% through enhanced hardware acceleration
- Implemented rigorous JWT/RBAC systems for 5+ enterprise-level admin panels, securing fleet management operations

Technical Projects

MOST — Hospitality Platform — Designed modular NestJS services for scalable AI travel workflows and architected an AI recommendation engine using LangChain and OpenAI

Tech Stack: NestJS · MongoDB · LangChain · OpenAI · AWS · Python · FastAPI

- Designed RAG pipelines and semantic search workflows using LangChain and vector embeddings to dynamically populate the UI with personalized travel recommendations
- Architected a robust communication workflow utilizing AWS S3, SES, SNS, and Pinpoint for transactional pipelines, and integrated Stripe for payment processing
- Owned production deployment of AI orchestration services and LLM inference pipelines, optimizing system throughput and reducing API response latency by 35%

DMS — Device Management System — Reverse-engineered proprietary camera network protocols to construct a high-concurrency Node.js streaming service

Tech Stack: Node.js · Docker · GCP · ffmpeg · MongoDB · Linux

- Reverse-engineered proprietary browser-based camera streaming protocols by analyzing obfuscated frontend websocket logic, enabling direct backend stream processing without vendor SDK support
- Architected a high-concurrency Node.js streaming service to capture feeds with audio via ffmpeg, buffering streams in RAM and flushing 10-second segments to disk
- Scaled the system to handle 2500+ concurrent cameras across 1000+ transit buses, delivering real-time telemetry with 99.9% uptime

Education

National Institute of Technology, Warangal 2022 – 2024
M.Sc. Mathematics & Scientific Computing CGPA: 8.0/10.0

Deshbandhu College, University of Delhi 2019 – 2022
B.Sc. (Hons.) Mathematics CGPA: 9.0/10.0

Technical Skills

Languages: JavaScript, TypeScript, Python, SQL
AI & Automation: LangChain, OpenAI API, Gemini API, RAG Pipelines, Vector Embeddings, LLM Workflows
Backend: Node.js, NestJS, Express.js, REST APIs, WebSockets, JWT, RBAC
Cloud & Infra: Docker, GCP, AWS, GitHub Actions, Linux
Databases: MongoDB, Redis, MySQL

Achievements

- Competitive Programming Link** : Top 100 Institute Rank on GeeksforGeeks with 534+ problems solved
- LeetCode Link** : 400+ Data Structures & Algorithms problems across all difficulty levels
- JAM 2022:** All India Rank 317 in Joint Admission Test for M.Sc. Mathematics (15,000+ candidates)